Linear Algebra II Midterm I Solution

1. Let V be an n-dimensional vector space and $I_V : V \to V$ be a linear transformation defined by $I_V(\boldsymbol{v}) = (\boldsymbol{v})$, $\mathbf{O}_V : V \to V$ be a linear transformation defined by $\mathbf{O}_V(\boldsymbol{v}) = \mathbf{0}_V$. Let $T : V \to V$ be a linear transformation such that $T^2 + T - 6I_V = \mathbf{O}_V$.

(a) Show that $\mathrm{Ker}(T+3I_V) \oplus \mathrm{Ker}(T-2I_V) = V$ and $\mathrm{Ker}(T+3I_V) = \mathrm{Im}(T-2I_V)$.

(b) Suppose $\langle\ ,\ \rangle$ is an inner product on V such that $\langle T(\boldsymbol{u}), \boldsymbol{v}\rangle = \langle \boldsymbol{u}, T(\boldsymbol{v})\rangle$. Show that $\mathrm{Ker}(T+3I_V) = \mathrm{Ker}(T-2I_V)^{\perp}$.

(c) Show that T is invertible. Express T^{-1} in terms of T and I_V.

(d) Suppose $n = 2$ and $\dim\mathrm{Ker}(T+3I_V) = 1$. Let $\boldsymbol{v} \in \mathrm{Ker}(T+3I_V)$ be a non-zero vector and a non-zero vector $\boldsymbol{u} \notin \mathrm{Ker}(T+3I_V)$ such that $\boldsymbol{v} = (T-2I_V)(\boldsymbol{u})$. Let $\beta = \{\boldsymbol{u}, \boldsymbol{v}\}$ be a basis of V.

(i) Find $\left[(T-I_V)^{-1}\right]_{\beta}$.

(ii) Hence express $(T-I_V)^{-1}(\boldsymbol{u}+T(\boldsymbol{v}))$ in terms of $\boldsymbol{u}$ and $\boldsymbol{v}$.

Solution

(a) Since $\mathrm{Ker}(T+3I_V)$ and $\mathrm{Ker}(T-2I_V)$ are subspace of V, we have $\mathrm{Ker}(T+3I_V) + \mathrm{Ker}(T-2I_V) \subset V$.

For any $\boldsymbol{v} \in V$, we have

$$5\boldsymbol{v} = (T+3I_V)(\boldsymbol{v}) - (T-2I_V)(\boldsymbol{v})$$
$$\boldsymbol{v} = (T-2I_V)(-\tfrac{1}{5}\boldsymbol{v}) + (T+3I_V)(\tfrac{1}{5}\boldsymbol{v})$$

Since $(T+3I_V)(T-2I_V) = T^2 + T - 6I_V = \mathbf{O}_V$,

$$(T+3I_V)\left[(T-2I_V)(-\tfrac{1}{5}\boldsymbol{v})\right] = \mathbf{0}_V,\quad \left[(T-2I_V)(-\tfrac{1}{5}\boldsymbol{v})\right] \in \mathrm{Ker}(T+3I_V)$$
$$(T-2I_V)\left[(T+3I_V)(\tfrac{1}{5}\boldsymbol{v})\right] = \mathbf{0}_V,\quad \left[(T+3I_V)(\tfrac{1}{5}\boldsymbol{v})\right] \in \mathrm{Ker}(T-2I_V)$$

Therefore, $V \subset \mathrm{Ker}(T+3I_V) + \mathrm{Ker}(T-2I_V)$. Hence $\mathrm{Ker}(T+3I_V) + \mathrm{Ker}(T-2I_V) = V$.

Let $\boldsymbol{u} \in \mathrm{Ker}(T+3I_V) \cap \mathrm{Ker}(T-2I_V)$. We have

$$T(\boldsymbol{u}) + 3\boldsymbol{u} = \mathbf{0}_V = T(\boldsymbol{u}) - 2\boldsymbol{u}$$
$$-3\boldsymbol{u} = T(\boldsymbol{u}) = 2\boldsymbol{u}$$
$$\boldsymbol{u} = \mathbf{0}_V.$$

That means $\operatorname{Ker}(T+3I_V) \cap \operatorname{Ker}(T-2I_V) = \{\mathbf{0}_V\}$. Hence

$$\operatorname{Ker}(T+3I_V) \oplus \operatorname{Ker}(T-2I_V) = V \text{ and}$$
$$\dim \operatorname{Ker}(T+3I_V) = n - \dim \operatorname{Ker}(T-2I_V).$$

Since $(T+3I_V)(T-2I_V) = T^2 + T - 6I_V = \mathbf{0}_V$, $\operatorname{Im}(T-2I_V) \subset \operatorname{Ker}(T+3I_V)$.

Furthermore, $\dim \operatorname{Im}(T-2I_V) = n - \dim \operatorname{Ker}(T-2I_V) = \dim \operatorname{Ker}(T+3I_V)$. We have

$$\operatorname{Im}(T-2I_V) = \operatorname{Ker}(T+3I_V).$$

(b) Let $\boldsymbol{u} \in \operatorname{Ker}(T+3I_V)$. For any $\boldsymbol{v} \in \operatorname{Ker}(T-2I_V)$,

$$\langle T(\boldsymbol{u}), \boldsymbol{v} \rangle = \langle \boldsymbol{u}, T(\boldsymbol{v}) \rangle$$
$$\langle -3\boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{u}, 2\boldsymbol{v} \rangle$$
$$-3\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 2\langle \boldsymbol{u}, \boldsymbol{v} \rangle$$
$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0$$

$\operatorname{Ker}(T+3I_V) \subset \operatorname{Ker}(T-2I_V)^{\perp}$. Since

$$\dim \operatorname{Ker}(T-2I_V)^{\perp} = n - \dim \operatorname{Ker}(T-2I_V) = \dim \operatorname{Ker}(T+3I_V),$$

we have $\operatorname{Ker}(T+3I_V) = \operatorname{Ker}(T-2I_V)^{\perp}$.

(c) $T(T+I_V) = (T+I_V)T = 6I_V$

$T\left[\frac{1}{6}(T+I_V)\right] = \left[\frac{1}{6}(T+I_V)\right]T = I_V$

T is invertible and $T^{-1} = \frac{1}{6}(T+I_V)$.

(d)(i) $[T]_\beta = \begin{pmatrix} 2 & 0 \\ 1 & -3 \end{pmatrix}$ (2)

$[T - I_V]_\beta = \begin{pmatrix} 2 & 0 \\ 1 & -3 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -4 \end{pmatrix}$ (1)

$\left[(T - I_V)^{-1}\right]_\beta = \begin{pmatrix} 1 & 0 \\ 1 & -4 \end{pmatrix}^{-1} = -\frac{1}{4}\begin{pmatrix} -4 & 0 \\ -1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 4 & 0 \\ 1 & -1 \end{pmatrix}$ (1)

(ii) $\left[(T - I_V)^{-1}(\boldsymbol{u} + T(\boldsymbol{v}))\right]_\beta = \left[(T - I_V)^{-1}(\boldsymbol{u} - 3\boldsymbol{v})\right]_\beta$

$= \frac{1}{4}\begin{pmatrix} 4 & 0 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ (1)

$= \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$(T - I_V)^{-1}(\boldsymbol{u} + T(\boldsymbol{v})) = \boldsymbol{u} + \boldsymbol{v}$ (1)

2. Let $M_{n\times n}(\mathbb{R})$ be the vector space of $n \times n$ matrix with real entry. Let $A \in M_{n\times n}(\mathbb{R})$ such that $Tr(A) \neq 0$. Define a mapping $T : M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R})$ by $T(M) = Tr(A)M - Tr(M)A$.

(a) Show that T is linear.

(b) Show that Ker(T) = span$\{A\}$ and Im(T) = Ker(Tr).

(c) Let $B \in M_{n\times n}(\mathbb{R})$ such that Tr(B) = 0. Find a matrix $M \in M_{n\times n}(\mathbb{R})$ such that $T(M) = B$ and Tr(M) = 2025.

Solution

(a) $T(\alpha M_1 + \beta M_2) = Tr(A)(\alpha M_1 + \beta M_2) - Tr(\alpha M_1 + \beta M_2)A$

$= \alpha Tr(A)M_1 + \beta Tr(A)M_2 - \alpha Tr(M_1)A - \beta Tr(M_2)A$

$= \alpha\left(Tr(A)M_1 - Tr(M_1)A\right) + \beta\left(Tr(A)M_2 - Tr(M_2)A\right)$ (2)

for all $\alpha, \beta \in \mathbb{R}$ and all $M_1, M_2 \in M_{n\times n}(\mathbb{R})$.

(b) $T(\alpha A) = Tr(A)\alpha A - Tr(\alpha A)A = \alpha Tr(A)A - \alpha Tr(A)A = 0_{n\times n}$. $\therefore$ $span\{A\} \subset Ker(T)$. (1)

Suppose $T(M) = 0_{n\times n}$. Then $Tr(A)M = Tr(M)A$ and $M = \frac{Tr(M)}{Tr(A)}A$. $\therefore$ $span\{A\} \supset Ker(T)$. (1)

$Tr(T(M)) = Tr\left(Tr(A)M - Tr(M)A\right) = Tr(A)Tr(M) - Tr(M)Tr(A) = 0$ for all $T(M) \in$ Im(T). (1)

$\therefore$ Im(T) $\subset$ Ker(Tr).

By Dimension Theorem,

$$\dim \mathrm{Im}(T) = \dim M_{n\times n}(\mathbb{R}) - \dim \mathrm{Ker}(T) = n^2 - \dim span\{A\} = n^2 - 1.$$

$$\dim \mathrm{Ker}(\mathrm{Tr}) = \dim M_{n\times n}(\mathbb{R}) - \dim \mathrm{Im}(\mathrm{Tr}) = \dim M_{n\times n}(\mathbb{R}) - \dim \mathbb{R} = n^2 - 1.$$

Hence, $\mathrm{Im}(T) = \mathrm{Ker}(\mathrm{Tr})$.

(c) $B = T(M) = Tr(A)M - Tr(M)A$

$T(B) = T^2(M) = Tr(A)T(M) - Tr(T(M))A = Tr(A)T(M)$

$B - Tr(A)M \in \mathrm{Ker}(T) = \mathrm{span}\{A\}$

$B - Tr(A)M = cA$

$Tr(B) - Tr(A)Tr(M) = cTr(A)$

$c = -2025$

$Tr(A)M = B + 2025A$

$$M = \frac{1}{Tr(A)}(B + 2025A)$$